

COMP 424 - Artificial Intelligence

Fall 2026

Lecture 1: Aug 31st, 2026

Course Info and General AI Discussion

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About Us

- Su Lin Blodgett
 - Visiting researcher at Mila
 - Researcher at Microsoft Research Montréal, 2020 – 2026
 - **Research topics:** natural language processing, responsible NLP and AI, evaluation and measurement
- Jackie CK Cheung
 - Associate professor, at McGill since 2015
 - Core member of Mila
 - **Research topics:** natural language processing, text summarization and generation, computational pragmatics, evaluation of NLP systems

What is AI to you?

- When you hear the words “artificial intelligence”, what do you think of?

Do these seem like AI to you?

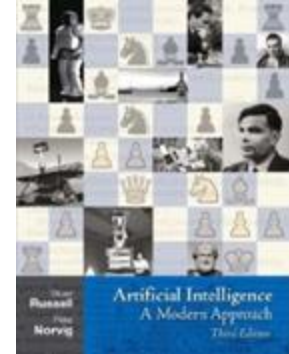
- ChatGPT
- Robot vacuum
- Tic-tac-toe playing bot
- Chess bot
- NPC (Non-playable character) in a video game with scripted interactions
- Calculator app
- GPS navigation
- Search engine

Let's come back to this, after the course intro!

About the Course

Textbook **(Recommended)**:

Russell & Norvig. “Artificial Intelligence:
A Modern Approach”, 4th ed.



We consider critical algorithms at the core of AI reasoning systems, of both historical and modern importance. Our focus is on AI for deliberative tasks including game playing, as well as probabilistic modeling and inference.

This cannot be a comprehensive coverage of everything called AI today due to scope – Machine Learning, Vision, Robotics and RL are specifically excluded due to other McGill courses.

Course Topics of COMP 424

- Uninformed Search
 - Informed search
 - Constraint satisfaction
 - Partial information and stochastic search
 - AI for 2 player games
 - Tree-based game solvers
 - Probabilistic reasoning
 - Learning probabilistic models
 - Causal probabilistic models
 - Reasoning with utilities
 - Sequential reasoning and decision-making
 - Responsible AI
- Warm up from 251
- Classical AI topics
- Game AI, project topic
- Probabilistic representations
- Utility theory

General Communication

- Su Lin's office hours: Mondays, time and location TBD
- Jackie's office hours: Wednesdays, 9:30 – 11am in MC 108N
- TAs will have office hours too (times and locations TBD)
- Use Ed discussion board for questions on content, technical queries, timing, etc.
- Use e-mail for personal issues

Evaluation

1. Individual assignments (20%)
2. Midterm (15%)
3. Final examinations (40%)
4. Final project: Implementation, check-ins, and written report (25%)

Assignments (20%)

- Four, roughly evenly spaced through term.
- Analysis or problem solving related to course content
- (Minimal) programming, problems, calculations
 - When coding needed, will be flexible on language

Examinations

- Midterm (15%) + Final (40%)
- Midterm date TBD; awaiting booking
- Final exams to be centrally scheduled, in person during exam period
- Mix of multiple-choice and written answer
- Closed-book but crib-sheet allowed

Course Project (25%)

- Design and implement a **game playing program**.
- Use ideas from the course to design your algorithm.
- Basic code (for a random player) will be provided
- Language of agent must be Python
- Evaluation: results, ongoing check-ins, written report
- Two rounds of performance evaluation with opportunities to have feedback and refine your solution

Course Policies

- **Course work is individual unless otherwise stated.** It is the responsibility of each student to understand the policy for the course and ask questions of the instructor if this is not clear.
- It is the responsibility of each student to **carefully acknowledge all sources** (papers, code, books, websites, individual communications) using appropriate referencing style when submitting work.

Collaboration Guidelines

- You are free to meet with fellow students(s) and discuss assignments with them. Writing on a board or shared piece of paper is acceptable during the meeting; however, you should not take any written (electronic or otherwise) record away from the meeting.
- This applies when the assignment is supposed to be an individual effort or whenever two teams discuss common problems they are each encountering (inter-group collaboration).

Collaboration Guidelines

- To assure that all collaboration is on the level, you must always write the name(s) of your collaborators on your assignment. This also applies when two groups collaborate.
- Suggestion: After the meeting, engage in a half hour of mind-numbing activity (like watching an episode of *The Office*), before starting to work on the assignment. This will assure that you are able to reconstruct what you learned from the meeting, by yourself, using your own brain.

Use of Generative AI

- Generative AI may be used in an assistive manner
- Acceptable use examples:
 - Help understand course content
 - Search for relevant information
 - Help brainstorm ideas
 - Check for grammar and style errors in writing and in code
- Unacceptable use examples:
 - As primary means of completing assignments
 - As primary means of completing project
- Use must be declared and referenced in your submission!

Automated Plagiarism Detection

- **We may use automated systems to detect possible cases of text or software plagiarism.** Cases that warrant further investigation will be referred to the university disciplinary officers. Students who have concerns about how to properly use and acknowledge third-party software should consult the course instructor or TAs.
- Unfortunately, we often detect and send cases of plagiarism to the office. This may delay your graduation! Do not do it! We have AI techniques at our disposal!

Acknowledgements

- Course slides based on previous offerings of COMP 424:
 - Doina Precup
 - Joelle Pineau
 - Jackie CK Cheung
 - Dave Meger
 - Golnoosh Farnadi
 - Prakhar Ganesh

What is Intelligence?

Definition of Intelligence?

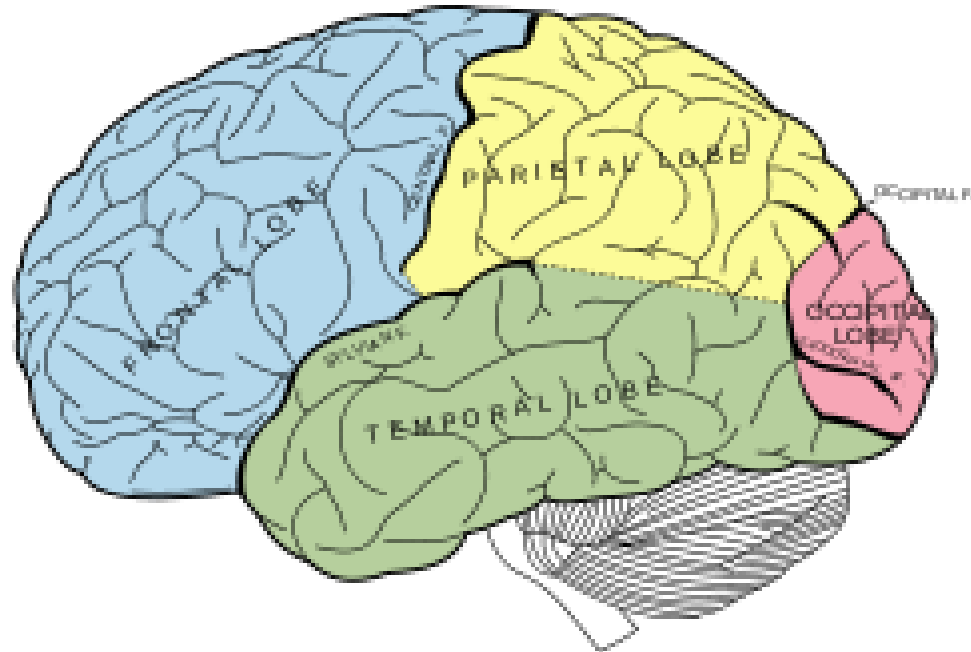
- Definition is contested. May be multi-factorial – ask a psychologist or cognitive scientist!

Possible characteristics and aspects of intelligence:

- Acquire, retain, and apply knowledge
- Apply logic and reason
- Be able to change and manipulate one's environment
- Be able to adapt, foresee, and plan
- Be able to deal with uncertainty
- Possible modalities of intelligence:
 - Spatial, verbal, logical, musical, social, ...

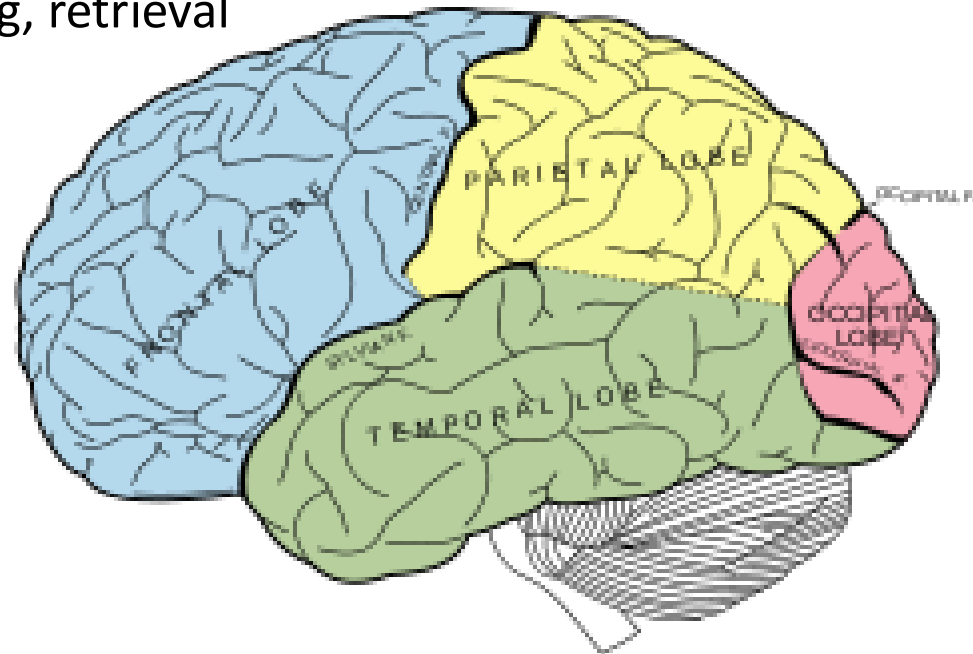
One Model: Biological Intelligence

- Sensory processing:
 - Visual cortex
 - Auditory cortex
 - Somatosensory cortex
- Motor cortex
- Cognitive functions:
 - Memory
 - Reasoning
 - Executive control
 - Learning
 - Language



Characteristics of Biological Intelligence

- A mix of domain-general and domain-specific capabilities
- **Domain-general:**
 - Memory formation, updating, retrieval
 - Learning new tasks
- **Domain-specific:**
 - Recognizing visual patterns
 - Recognizing sounds
 - Learning language



What is AI?



What is AI?

AGI could arrive within a year, researcher says

Robots are like children — one day they will make us proud

Tech leaders are nervous about advances in artificial intelligence but we are getting nearer to the moment when their capabilities astound us

What is AI?

Human intelligence:

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Artificial Intelligence:

- Computer vision
- Signal/speech processing
- Haptics
- Robotics
- Knowledge representation
- Search, inference
- Planning, decision-making
- Model learning
- Language understanding

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Possible Goals of AI

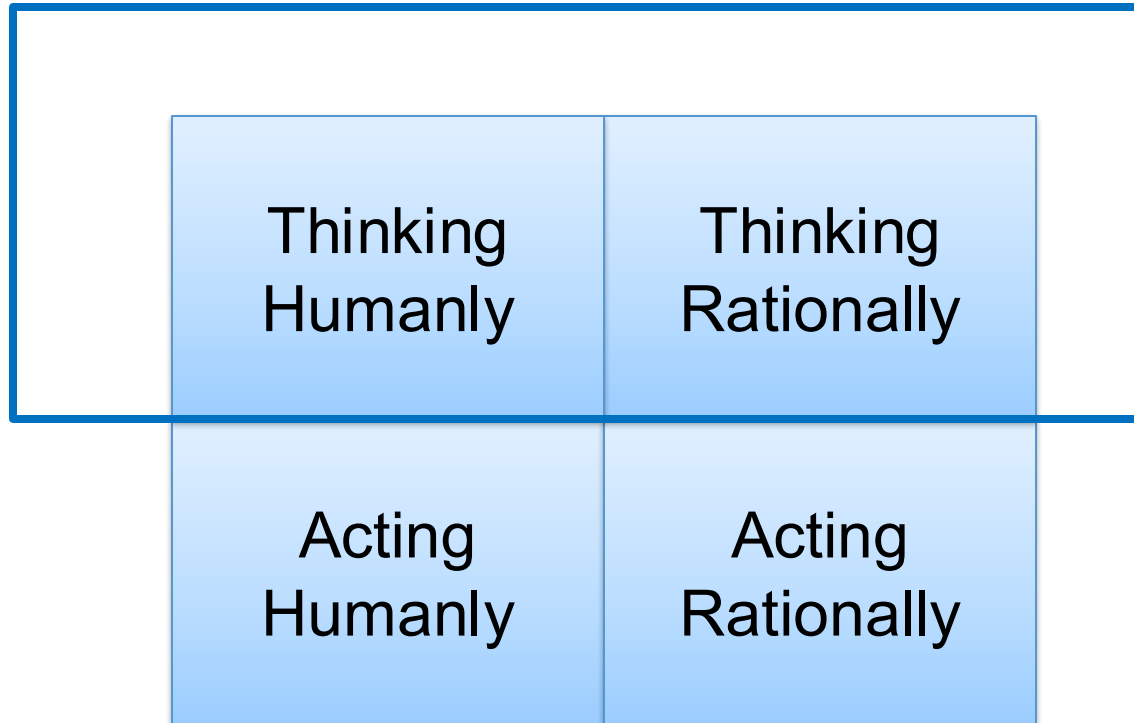
- Modeling or replicating human cognition using computers
- Replicating human behaviours using computers
- Studying problems that others don't know how to solve
 - Cool stuff!
 - Game playing, machine learning, data mining, speech recognition, computer vision, web agents, robots
 - Useful stuff!
 - Medical diagnosis, fraud detection, genome analysis, object identification, space shuttle scheduling, information retrieval

Goals of AI

Thinking Humanly	Thinking Rationally
Acting Humanly	Acting Rationally

Goals of AI

Let's first consider these boxes



Potential Issues

- What does it mean to “think humanly”?
 - Studied in cognitive science
 - Can use computational methods to model and study this!
 - Not the focus of this course!
 - Also: can an AI have a mind and consciousness? No agreed definition and test!
- What does it mean to “think rationally”?
 - Possible answer: use a well-understood and described logic system.
 - Is this the right goal for developing intelligent systems?

Goals of AI

What about this?



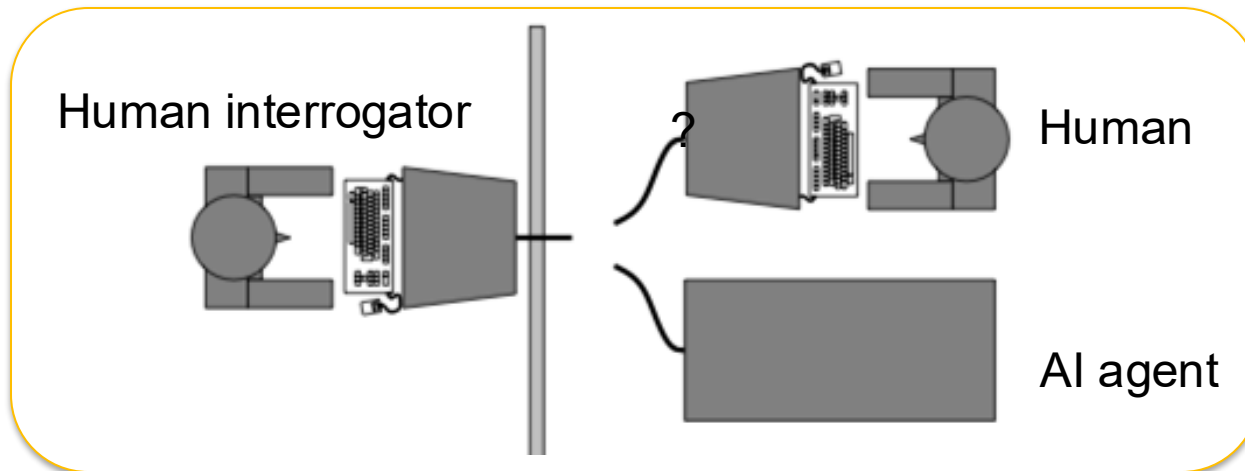
Thinking Humanly	Thinking Rationally
Acting Humanly	Acting Rationally

Acting Humanly

- AI is about duplicating what the (human) brain DOES.
- Alan Turing (1912-1954) had interesting thoughts about this.

Can a machine think? -> If it could, how would we tell?

Turing (1950): "Computing machinery and intelligence"



An operator interacts with either the human or the AI agent.
Can he correctly guess which one?

Turing's Prediction

- By 2000, a machine would have a 30% chance of fooling a lay person for 5 minutes.
 - This actually happened in 2014:
<http://www.bbc.com/news/technology-27762088>
 - Does this mean that we have solved AI?
- Suggested major components of AI:
 - Knowledge representation, automated reasoning, language understanding, machine learning

AI = Acting Humanly?

- Humans have biological resource constraints
 - Limited memory, thinking speed, attention span...
- Humans are not necessarily the gold standard!
- Humans are often “irrational”
 - Or else we do not understand our own notion of rationality
 - e.g., don't act towards our own goals; hold contradictory beliefs or preferences
 - Kahneman and Tversky demonstrate ways that people are systematically irrational.

Different Goals of AI

Thinking Humanly	Thinking Rationally
Acting Humanly	Acting Rationally

Let's try this!



Acting Rationally

- Rational behaviour = doing the “right” thing.
- Doing what is expected to **maximize goal achievement**, given the **available information** and **available resources**.

This is the flavour of AI we will focus on.

Who gets to decide what is the “right” thing?
– A question to ponder.

Finally, Our Working Definition of AI

Developing models and algorithms that can produce rational behaviours in response to incoming stimuli and information.

Does this change your views?

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Different Goals of AI

Thinking Humanly	Thinking Rationally
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Rational Agents

- This course is about designing **rational agents**.
 - An agent is an entity that **perceives** and **acts**.
 - Goal: Learn a function mapping *percept histories* to *actions*:
$$f : P^h \rightarrow A$$
- A rational agent implements this function such as to maximize performance.
 - Performance measures: goal achievement, resource consumption, ...
- **Caveat:** Resource constraints (time, space, energy, bandwidth, ...) which make perfect rationality unachievable
- **Objective:** Find best function for given information and resources

Note on Terminology

- What terms have we used that could be applied to both AI and humans?
 - E.g., “rational”
- What assumptions may differ when applying the same term to the two?

What is AI?

AGI could arrive within a year, researcher says

Robots are like children — one day they will make us proud

Tech leaders are nervous about advances in artificial intelligence but we are getting nearer to the moment when their capabilities astound us

How is "AI" being used?

AGI could arrive within a year, researcher says

Robots are like children — one day they will make us proud

Tech leaders are nervous about advances in artificial intelligence but we are getting nearer to the moment when their capabilities astound us

AI Beginnings

- Pre/early compute:
 - Feedback devices enable water supply, steam power, encryption and code breaking
 - Devices provide automation not computation. Can never surpass creator's understanding
- ENIAC: First super-computer, created in 1946.
- Early work in 1950s:
 - Rosenblatt's perceptron
 - Samuel's checkers player
- AI successes come in many waves

Dartmouth Conference (1956)

"We propose that a 2 month, 10 man study of artificial intelligence be carried out during the summer of 1956 at Dartmouth College in Hanover, New Hampshire. The study is to proceed on the basis of the conjecture that every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it. An attempt will be made to find how to make machines use language, form abstractions and concepts, solve kinds of problems now reserved for humans, and improve themselves."

Dartmouth Conference (1956)

- Some of the attendees:
 - **John McCarthy**: LISP, time-sharing, application of logic to reasoning
 - **Marvin Minsky**: popularized neural networks and showed their limits, introduced slots and frames
 - **Claude Shannon**: information theory, juggling machine
 - **Allen Newell** and **Herb Simon**: bounded rationality, general problem solver, SOAR
- The meeting coined the term “**artificial intelligence**”



Early AI Hopes and Dreams

- Make programs that exhibit similar signs of intelligence as people: **prove theorems, play chess, have a conversation.**
- Logical reasoning was key.
- Learning from experience was considered important.
- The research agenda was geared towards *building general problem solvers*.
- There was a lot of hope that natural language could be easily understood and processed.

AI Downswings

- Early successes did not scale up!
 - Demos were impressive, but only worked in a narrow domain
 - e.g., Machine translation for general texts

1966	Perceived failure of machine translation
1973, 1974	Major cut in AI research funding
1987-1993	“AI Winter”: many companies working in AI fail

- Much progress actually made in this period, but overpromising of results led to disappointments

Recent AI: Statistical Techniques

- Heavy use of probability theory, decision theory, statistics.
- Trying to solve **specific problems** rather than aim for general reasoning.
- AI today is a collection of sub-fields:
 - Perception and computer vision.
 - Natural language understanding.
 - Robotics
 - Etc.
- Deep learning unleashed a further wave of enthusiasm about AI progress (especially 2014 onwards)
- Recent claims that large language models and multi-modal models surpass the problem-specific limitation and are moving towards “general” intelligence or even “superintelligence”.

Next Class

- Start with discussion of **search** as a fundamental paradigm for AI!
- **Starting next week (Sept 9), we'll be in MCMED 522.**